TEAM:	#:

☐ Single-Occupant ☐ Multi-Occupant

Driver / Passenger Registration	Sticker & Initials	Electrical	Sticker & Initials	Solar Collector	Sticker & Initials
Notes:		Notes:		Notes:	
Driver Operations	Sticker & Initials	Battery Protection System	Sticker & Initials	Safety	Sticker & Initials
Notes:		Session A Notes: Session B Notes:		Notes:	
Lights & Vision	Sticker & Initials	Mechanical	Sticker & Initials	Impound / MOV	Sticker & Initials
Notes:		Notes:		Impound Method ☐ In Vehicle ☐ External Box MOV Metered Charging ☐ Certified ☐ Not Certified	
Body & Sizing	Sticker & Initials	Dynamics	Sticker & Initials	ASC Support	Sticker & Initials
Notes:		Notes:		Notes:	

FSGP (Track)	Passed-HQ Receipt Date/Time/Initials	ASC (Road)	Passed-HQ Receipt Date/Time/Initials

Station	Regulation	Scrutineering Penalty Issue Description	Penalty Value (Per Day)
			Miles ____________ Laps ____________
			Miles ____________ Laps ____________
			Miles ____________ Laps ____________
			Miles ____________ Laps ____________
			Miles ____________ Laps ____________

TEAM:	**#:**

Driver Name / Regulation	**Driver 1**	**Driver 2**	**Driver 3**	**Driver 4**
11.1.A – Driver is registered with HQ (has ID), is 18 or older with valid DL				
9.9.C – Common Ballast (Only an option for SOVs)	Mass [kg] ________________		Ballast Tag # ________________	
11.2 – Driver Mass [kg] (includes driving clothes and shoes but not helmet)				
9.9 – Ballast Mass [kg] – ballasted to 80 kg (176 lb)				
Wristband Color	Orange Yellow Green Blue Purple	Orange Yellow Green Blue Purple	Orange Yellow Green Blue Purple	Orange Yellow Green Blue Purple
Wristband ID #				
Ballast Security Tag ID #				

11.1.A.2 **# of solar car drivers registered** – (2 min, 4 max)		
11.2.B **Helmets** – Type/Rating – Snell, FIA, DOT FMVSS (3/4” min padding), ECE, AS/NZS		
11.2.C **Shoes** – Solid sole, closed-toe		

***** FOR MULTI-OCCUPANT VEHICLES, COMPLETE PAGE 2 FOR PASSENGERS *****

Sticker & Initials

Station Manager: ____________________

Entrance: All occupants report with ballast material, helmet(s), proper driver / passenger uniforms

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	#:

***** FOR MULTI-OCCUPANT VEHICLES *****

Passenger Name / Regulation	**Passenger 1**	**Passenger 2**	**Passenger 3**	**Passenger 4**
11.1.B – Registered with HQ (has ID), is 18 or older				
11.2 – Passenger Mass [kg] (includes clothes and shoes but not helmet)				
9.9 – Ballast Mass [kg] – ballasted to 80 kg (176 lb)				
Passenger Number Punched (1-8, X)	1 2 3 4 5 6 7 8 X	1 2 3 4 5 6 7 8 X	1 2 3 4 5 6 7 8 X	1 2 3 4 5 6 7 8 X
Wristband ID #				
Ballast Security Tag ID #				

Passenger Name / Regulation	**Passenger 5**	**Passenger 6**	**Passenger 7**	**Passenger 8**
11.1.B – Registered with HQ (has ID), is 18 or older				
11.2 – Passenger Mass [kg] (includes clothes and shoes but not helmet)				
9.9 – Ballast Mass [kg] – ballasted to 80 kg (176 lb)				
Passenger Number Punched (1-8, X)	1 2 3 4 5 6 7 8 X	1 2 3 4 5 6 7 8 X	1 2 3 4 5 6 7 8 X	1 2 3 4 5 6 7 8 X
Wristband ID #				
Ballast Security Tag ID #				

11.1.B.1 **# of solar car passengers registered** – (8 max)		

TEAM:	**#:**

Regulation	10.3.A.15 **Roll Cage Clearance –** 50 mm min roll cage to helmet, 30 mm min padding to helmet	9.8.A **Egress** – Buckled in seat to standing clear of solar car, no wheel chocks, unassisted	
		Primary (< 10 sec)	Secondary (< 15 Sec) – MOV Only
Driver 1			
Driver 2			
Driver 3			
Driver 4			
Passenger 1			
Passenger 2			
Passenger 3			
Passenger 4			
Passenger 5			
Passenger 6			
Passenger 7			
Passenger 8			

Regulation	**Grade**	**Comments**
11.4 **Water/Fluids** – Water/fluid provision (2L min / per occupant), secured, one hand access		
11.5 **Radios/Communication** – Driver in radio contact with team, hands free (not a cell phone)		
9.9.B **Ballast Carriers** – One per occupant, securely mounted within 300 mm of hip point		
9.9.D **Ballast Access** – Ballast bag/ID tags visibly accessible for Official verification		
9.9.C **Common Ballast Box (SOV Only)** – Securely mounted and sealable if equipped		☐ Common Ballast ☐ Box Sealed ☐ No Common Ballast

Sticker & Initials

Station Manager: ____________________

Entrance: All occupants report with ballast, helmet(s), proper driver / passenger uniforms with fully assembled solar car and radio communication;
Yellow status or better in Driver Registration

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

Regulation	Grade	Comments
Vision		
9.7.D **Forward/Side Vision** – 6.4 m above @ 12.2 m forward, ground @ 8 m forward, 100° side to side, 50 mm high letters @ 3m forward & @3m to each side		
9.7.E **Rear Vision** – Camera/view screen securely fixed in position & visible in normal driving position		
9.7.E.1 **Rear Vision** – 15 m rearward, 30° L/R off center, single reflex image		
Lighting/Signals		
9.4.A **Lighting – Front Turn** – Amber, visible 80° out, 45° in, 5° down, 15° up at 30 m		
9.4.G **Lighting – Side Marker** – Amber, visible 60° arc, from 5° to 65° off centerline (viewed from rear), 5° down, 15° up at 30 m		
9.4.B **Lighting – Brake** – Red, visible 45° L/R, 5° down, 15° up at 30 m		
9.4.C **Lighting – Rear Turn** – Red/amber, visible 80° out, 45° in, 5° down, 15° up at 30 m		
9.4.D **Lighting – High Mount Brake** – Red, visible 10° L/R, 5° down, 10° up at 30 m		
9.4.E **Lighting – BPS Fault** – White, visible 10° L/R, 10° up at 30 m		
9.4.F **Emergency Hazard Lights Format** – Front turn, side markers, rear turn flash in sync		
Horn		
9.5 **Horn** – 75-102 dB sound level @ 15 m forward, permanently mounted, steering wheel operated, up to 5 min continuous operation		

Sticker & Initials

Station Manager: ____________________

Entrance: Driver in fully assembled solar car, radio communication, & Battery Spill Kit;
Yellow status or better in Driver Registration & Electrical

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

Regulation	**Grade**	**Comments**
Dimensions & Body		
9.1 **Solar Car Dimensions [m]** – Max dimensions: L = 5.8 m, W = 2.3 m, H = 1.65 m		L ___________ W ___________ H ___________
9.3 **Ground Clearance** – Greater than 75mm		
8.I.I & 9.2 **Operational Configuration** – Body remains fixed (no reorientation/tilting) when moving under its own power		
9.7.B & 9.7.C **Windshield** – securely mounted, clear, shatter resistant, method to clear rain, distortion free		
9.10.A **Solar Car Numbers** – High contrast color, min 50 mm background, min 250 mm H x 120 mm W numerals, min 40 mm solid brush stroke, min 25 mm spacing, visible from 3 m at 1.8 m above ground		
9.10.B **Institution Name** – Both sides of car, approved abbreviations, more prominent than any team sponsor logo/name, no disruptive or offensive graphics, visible from 3 m at 1.8 m above ground		
9.10.C **Event Logo** – Space (200 mm H x 500 mm W) on both sides of car, visible from 3 m @ 1.8 m above ground		
9.10.D **National Flag** – Displayed on both sides of car by windshield (min size 70 mm x 40 mm)		
9.10.E **Front Signage** – Host institution name, space for event logo (150 mm x 150 mm), unbroken area forward of windscreen, visible from top and front elevation view		
9.4.A **Front Turn Signals** – Located at front of car, min 600 mm apart (400 mm for cars narrower than 1300 mm), min 350 mm above ground		
9.4.G **Side Marker Turn Signals** – Located on each side of car, 500 to 1800 mm from absolute front of car, within 400mm from extreme outer edge		
9.4.B **Rear Brake Lights** – Located at rear of car, min 600mm apart (400 mm for cars narrower than 1300 mm), within 400 mm from extreme outer side edge, min 350 mm above ground		
9.4.C **Rear Turn Signals** – Located at rear of car, min 600mm apart (400 mm for cars narrower than 1300 mm), within 400 mm from extreme outer side edge, min 350 mm above ground		
9.4.D **High Mount Center Brake Light** – Rearward facing, located within 150mm of the highest fixed point of the car and above top of rear brake lights		
9.4.E **BPS Fault Indicator Light** – Located adjacent (left, right, top, or bottom) to the center brake light		
Front of Car to Driver's Headrest Distance [m] – (Used for upward vision calcs)		D ___________

TEAM:	#:

Regulation	Grade	Comments
Cockpit		
7.1.A, 10.3.B.1 **Single Occupant Class # of Seats** – (Max of 1)		
7.1.B, 10.3.B.2 **Multi-Occupant Class # of Seats** – (2-4)		
10.3.B.3 **Seating Position** – Seat within ±10° of forward facing		
10.3.B.4 **Back and Head Restraint** – 800 mm min hip point to head restraint top (750 mm min for MOV rear seats)		
10.3.B.5, 10.3.B.6 **Occupant heels below hip point** – >90° angle between shoulders, hips, knees		
10.3.C **Occupant Space Check**		
9.7.A **Visibility –** 700 mm min eye height from ground (all occupants)		
10.3.D **Belly Pan** – Full occupant isolation from road, able to support 80 kg, occupant torso and limbs above lower element of structural chassis, full free motion of occupant limbs unable to contact the road		
10.3.A.13 **Padding** – Roll cage padded around helmet meeting SFI-45.1/FIA 8857-2001 A or B or better, wrapping 50% min around roll cage member		
10.3.A.14 **Headrest** – Securely mounted headrest for each occupant with 19 mm min thick padding		
9.6 **Outside Air Circulation** – Cockpit vents directed at occupant's face / fan present if intake is from wheel openings		
9.8.B **Egress** – Can be opened from both inside and outside, no tape used at egress point, positive latch, release located within 300 mm from opening edge		
9.8.B.4 Egress openings and external release instructions clearly marked		
Vehicle Mass & Tires		
Vehicle Mass – Ballasted occupants in all seats (multiply kg value by 2.205 to get lb)		LF [kg] ___________ RF [kg] ___________ LR [kg] ___________ RR [kg] ___________ Total [kg] ___________ Total [lb] ___________
10.2.A.2 **Wheel/Tire Sets** – Left-most to right-most contact patches at least half overall car width		
10.2.D **Tire Speed & Load Ratings (per tire)** – Tires inflated within manufacturer's rating, tube-type tires need tubes, US DOT or UNECE		Speed Symbol _____ Max Speed [MPH] _____ Load Index _____ Max Load [kg] _____ Load Range _____ Max Inflation [PSI] _____
10.2.E **Wheel/Rim** – Profile matches bead requirements of tire		
Tire Set Configuration Notes:		

TEAM:	**#:**

Regulation	Grade	Comments
8.1 **Solar Collector Sizing**		
8.1.A **Cell Type**		Type ________________
8.1.B **Size –** SOV: $6m^2$, MOV: $5m^2$		
5.3.E **Solar Cell Technology** – Solar cells match information given in VDR		
8.1.F **Example Cell and Layout Map** – Matches physical solar collector on car		
8.1.E **No More than 3 Cell Types or Sizes Used**		
8.1.D **Concentrators** – Total aperture of the solar collector not exceeding total allowable area for non-concentrator photovoltaic solar collectors		
8.1.G **Solar Collector Connections & Stands** – Solar collector stands are self-supporting, all portions of the collector are carried by solar car (stands, supports, cables, electrical connections, etc)		

Sticker & Initials

Station Manager: ______________________________

Entrance: Driver and ballasted occupants in fully assembled solar car; Yellow status or better in Driver Registration

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

Regulation	Grade	Comments
8 **Power** – No power source aside from solar collector		
5.3.C & 8.2 **Storage Batteries** – Match submitted battery approval form (cell type, qty, pack config, etc)		# of Battery Cells in Vehicle Pack ______________
8.2 **Energy Storage System Information** Pack Capacity [kWh] – (5.25 kWh max for SOV, 15.5 kWh max for MOV) _____________ Pack Chemistry ☐ Li-S ☐ Li-Ion / Li Polymer ☐ LiFePo4 ☐ 8.2.B Other ___________________________________		
8.4.D **Battery Ventilation** – Whenever battery is enabled, forced ventilation pulls air from battery enclosure through sealed ducting to exterior vent (Fan can operate from supplemental if BPS trips)		
8.5 **External Cooling** – Not permitted unless powered by main battery or in an emergency		
8.4, 8.4.C **Battery Enclosure** – non-conductive inside, no more than 2 enclosures, 10mm labels on top		
8.4.A, **Isolation** – Greater than 100Ω/1V max battery potential from terminal to chassis		
8.7.C **External Power Cut Off Switch** – Location, marking, operation, covering, load rated		
8.10 **Electrical Shock Hazards** – HV > 60V protected and marked with 10 mm min high letters		
8.2.B **Other Storage Techniques** – Ultracapacitors, flywheels, fuel cells		
8.4.B **Battery Mounting** – Enclosure/modules secure		
8.2.C **Supplemental Batteries** – Must power BPS momentarily for startup checks – must power BPS, BPS strobe & BPS fault dash indicator during BPS fault – specify if it powers other allowable systems		☐Telemetry ☐Horn ☐Driver Ventilation Fans ☐Main Power Contactors
8.4 **Supplemental Pack Location** – Battery enclosure		
8.6.A **Main Fuse** – In series with batt, not exceeding 200% peak current draw or 75% of wire capacity		
8.6.B **Branch** – Proper fuses on wiring off main bus		
8.6.C **Voltage Taps** – In batt enclosure, <10mA limit		
8.7.A **Main Power Contactors** – In batt enclosure, non-latching, normally open, DC rated, able to interrupt peak current, location in circuit		
8.8 **Cable Sizing** – Proper size for system current		
8.9.B **Accelerator** – Free moving, return to zero, located right of brake pedal (if equipped)		
8.9.A **Control** – Driver has sole control		
8.9.C **Cruise Control** – Driver activated only, automatic deactivation by brake pedal/power off		
8.9.D **Reverse** – Under own power		
8.4.E **Security** – Official battery seal applied		

Sticker & Initials

Station Manager: ______________________________

Entrance: Fully assembled solar car & Battery Spill Kit

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

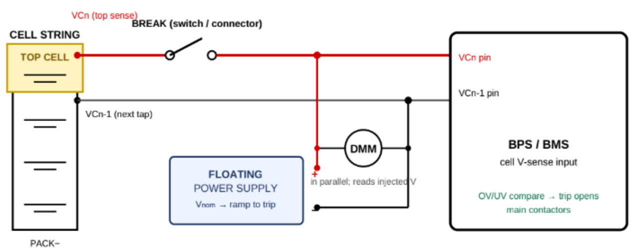
Reg 8.3 | OVER / UNDER VOLTAGE (OV & UV) Reg. 5.3.D.5 & 5.3.D.6 ☐ Pass ☐ Fail

Nominal Vnom : ________ V @_______ °C

Condition	Mfr Limit (cell/module)	BPS Setpoint	Measured Trip	Result	Notes
Over-Voltage (max)	_______ Vmax	______ Vmax_set	_______ Vmax_trip	☐ Pass ☐ Fail	
Under-Voltage (min)	_______ Vmin	______ Vmin_set	_______ Vmin_trip	☐ Pass ☐ Fail	

Pass: OV trip ≤ V max ; UV trip ≥ V min. Acceptable fault tolerance: measured trip within ± 0.01 V of setpoint (cell / module level only).

Example OV / UV Hook-Up: Break Top-Most Cell Sense; Inject with Floating Supply + DMM



Procedure: inject reference at top-most cell sense (OV/UV)
1. Read live top-cell V on DMM; verify test-point label and polarity.
2. Open the BREAK (switch or connector) in the top-cell sense lead.
3. Connect floating supply + DMM: + to BPS VCn_pin, - to VCn-1 tap; preset Vnom.
4. Reset BPS; hold Vnom 10s and confirm BPS stays active
5. Ramp up to OVtrip (<=0.01 V/step near setpoint); record. Return to Vnom, Reset
6. Ramp down to UVtrip; record. Bring back up to Vnom.

Notes

____

____

____

____

Reg 8.3 | OVER CURRENT (OC) Reg. 5.3.D.7 ☐ Pass ☐ Fail

Current Sensor Type & Test Requirement:

☐ **Shunt** **Ratio**: ___________**mV/A** **Gain amp needed?** ☐ **Yes** ☐ **No**	**Notes:** ____________________________ ____________________________	☐ **Hall Effect** / ☐ **Other** **Loop turns N**: ___ **Input I**: ____**A** **Output current trip**: _______**A**	**Notes:** ____________________________ ____________________________

Condition	Mfr Limit	BPS Setpoint(trip)	Shunt / Hall Effect Vsen_max/Iref	Measured Trip	Result
Over-Current (Charge) -Imax	______ **A**	_______ **A**		_____**A chrg_trip**	☐ **Pass** ☐ **Fail**
Over-Current (Discharge) +Imax	______ **A**	_______ **A**		_____**A dischrg_trip**	☐ **Pass** ☐ **Fail**

Pass: trip ≤ I max for each direction. Acceptable fault tolerance: measured trip within ± 0.2 A (≈ 2%) of setpoint.

OC Procedure: Record the sensor type and scaling: shunt mV/A ratio (with any amplifier gain) or hall-effect turns ratio. Simulate the trip current at the sensor instead of passing full pack current. For a shunt, inject the equivalent millivolt signal (setpoint × mV/A ratio) across the sense leads while reading it on the DMM. For a hall-effect sensor, pass a known current through N wire loops so that N × I equals the trip point. Confirm the BPS stays active just below the setpoint, raise the signal slowly until trip, and record the value. Repeat for charge and discharge. Each trip must occur at or below the manufacturer's Imax for that direction, open the main contactors, and latch until reset.

Reg 8.3 | OVER TEMPERATURE (OT) Reg. 5.3.D.4 ☐ Pass ☐ Fail

Condition	Mfr Limit	BPS Setpoint(trip)	Measure Trip	Result	Notes
Over-Temperature (Charge)	_______ °C	_______ °C	______ °C	☐ Pass ☐ Fail	
Over-Temperature (Discharge)	_______ °C	_______ °C	______ °C	☐ Pass ☐ Fail	

Pass: trip ≤ T max for each setpoint. Acceptable fault tolerance: measured trip within ± 2 °C of setpoint.

OT Procedure: Ensure the sensor's starting temperature is below both setpoints. Apply a controlled heat source to the BPS temperature sensor with a calibrated reference probe adjacent to it. Increase temperature gradually (≤1 °C/sec near the setpoint) and record the probe temperature when the charge trip occurs. Cool below the setpoint, reset the BPS, and repeat for the discharge trip. Each trip must occur at or below the manufacturer's Tmax, open the main contactors, and remain latched until manually reset after cooling.

Regulation	Grade	Comments
8.3.A.5 Fault Latch – BPS fault latches; manual driver clearing		
8.7.B Fault Dash Indicator – Illuminates on BPS trip		
9.4.E.2 BPS Fault Indicator Light – Illuminates in 3 sec		

Sticker & Initials

Station Manager: ______________________________________

Entrance: Fully assembled solar car, battery pack, BPS, & Battery Spill Kit; Yellow status or better in Electrical

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

Regulation	Grade	Comments
5.3.A **Mechanical VDR** – vehicle matches mechanical report		
8.4.B **Battery Enclosures** – structurally sound and properly secured to chassis		
9.9.B, 9.9.C **Ballast Boxes** – structurally sound and properly secured to chassis		
10.1 **Body Panels and Solar Collector** – securely fastened to prevent unintended movement		
10.1.C **Collector Attachment** – 2 independent methods		
10.2.A **Wheel Configuration Acceptable**		
10.2.B **Wheels** – Designed to meet application loads		
Remove a front and rear wheel to check fit up		
10.8 **Towing Hardpoint** – Accessible for forward towing with body in place (canopy can be removed)		
Occupant Cell		
10.1.A **Covers and Shields** – All moving parts covered to protect against accidental contact		
10.3 **Occupant Cell** – Designed for protection, will not cause undue strain		
10.7.A, Appendix A **Steering Wheel** – Sufficient strength steering wheel with continuous perimeter		
10.3.E, 10.3.E.3 **Safety Belts** – Valid (non-expired) commercial FIA or SFI compliant 5-point harness, properly attached to acceptable attachment points		
10.3.E.6, 10.3.E.7, 10.3.E.8 **Shoulder Belt Placement**		
10.3.E.6, 10.3.E.9 **Lap Belt Placement**		
10.3.E.6, 10.3.E.10 **Anti-Submarine Belt Placement**		
10.3.E.4 **Safety Belt Chafing Through Seat**		
10.3.A.1, 10.3.A.4 **Roll Cage** – Encompasses occupants from shoulders up, metallic elements		
10.3.A.2 **Structural Chassis** – Designed to encompass occupants in all directions, includes suspension mounts		
10.3.A.16 **Shatter Protection** – Composites near head		
10.5.E & 10.5.F **Pedal Placement** – Brake pedal activation, spacing between pedals		
8.9.B **Accelerator Pedal Placement** – Free moving, right foot activated, return to 0, right of the brake pedal		
Steering		
10.1.B **Clearance** – Moving parts are interference free		
10.1.B, 10.7.D **Steering Static Test** – Can turn lock to lock while still, no excessive play in steering		
10.7.B **Steering stops** – In place and functional		

TEAM:	**#:**

Regulation	**Grade**	**Comments**
Brakes		
10.5.F **Hand Brakes** – Lock to lock use without repositioning hands (if equipped)		
10.5.A **Brakes** – Dual independent and balanced co-reactive braking system		
10.5.B **Brake Pads** – Contact area > 6.0 cm^2, initial thickness >= 6.0 mm, full contact with rotor		
10.5.D **Brake Lines** – Appropriately sized and constructed		
10.5.G.2 **Mechanical Rear Brake Performance** – 15% of vehicle weight (if equipped)		15% of Vehicle Weight [lb] ________________ Pull Test [lb] ________________
10.5.G.3 (Or Other Areas) – **Volume Limiting Valve**		
10.6 **Parking Brake** – Functional parking brake (must hold 10% of vehicle weight in both directions), lockable, independent, non-tire contact style		10% of Vehicle Weight [lb] ________________ Pull Test [lb] ________________
Fasteners/Hardware		
10.4.C **Rod-Ends** – Secured with sufficiently torqued jam nuts (flex-loc/safety wire not required)		
10.4.F **Hub Nuts** – 10.9 mm thick threaded portion required for single central hub nut securing the wheel		

10.4.E **Critical Areas**	Steering	Brakes	Front Suspension	Rear Suspension	Seat/Safety Harness	Drive Train	Battery Enclosure	Ballast Box	Parking Brake
10.4 **Means of Retention** – Must not use friction, glue, or press fit as the only retention mechanism									
10.4.A **Bolts** – SAE grade 5, M 8.8 or AN/MS, two threads beyond nut, no shaved heads									
10.4.B **Securing Bolts** – Safety wire, cotter pins, flex-loc nuts, or other approved method (Nylock, Nordlock, Loctite, and thread distortion insufficient)									
10.4.D **Buckles & Straps** – No plastic luggage type buckles or single push release straps									
Fastener/Hardware Notes:									

Sticker & Initials

Station Manager: ____________________________

Entrance: Fully assembled solar car (will be disassembled as necessary within the station)

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

Regulation	Grade	Comments
U-Turn Test		
10.7.C **Turning Radius** – Tire contact patches remain within 15 m wide lane		Right Turn ____________ Left Turn ____________
Figure-8 Test		
10.2.A.1 **Tire and Wheel Requirements** – All wheel tires must remain on the ground		
10.1.B **Clearance** – Body and other stationary parts must not contact moving parts (except bearings)		
10.9 **Dynamic Stability** – Vehicle must exhibit sufficient stability during test		
10.9.A **Figure 8** – Vehicle must negotiate Figure-8 in < 8 sec per side without hitting cones		Time for Figure-8 [sec] ____________
Braking Test		
10.9 **Dynamic Stability** – Vehicle must exhibit sufficient stability during test		
10.5.C, 10.9.D **Braking Performance** – Vehicle must completely stop from >= 30 MPH (48 km/h) @ > 4.72 m/s^2 deceleration without undue veering left or right		Time [sec] ______ Starting Speed [MPH] ______
Slalom Test		
10.9 **Dynamic Stability** – Vehicle must exhibit sufficient stability during test		
10.9.C **Slalom** – Negotiate within in 11.5 sec		Time [sec] ______ Starting Speed [MPH] ______
Acceleration Test		
10.9.E **Accel** – 18 m within 8 sec from 0 MPH		Time [sec] ____________
High Speed Stability Test		
10.9.B **Lane Stability** – Vehicle must be able to stay within a 3.5 m wide lane up to 65 MPH (105 km/h) & exhibit sufficient stability and during test		Demonstrated Speed Capability [MPH] ____________

Before Passing Team…		
10.5.G.3 – Lock out proportioning valve		☐Yes ☐No
Measure tire pressure [PSI]		LF _______________ RF _______________ LR _______________ RR _______________
MOV: Test all driver / passenger configurations!		

Sticker & Initials

Station Manager: ______________________________

Entrance: All drivers & passengers report to station with fully assembled solar car, ballast, Battery Spill Kit, radio communication, & spare tires; Yellow status or better in Driver Registration, Driver Operations, Lights & Vision, Body & Sizing, Electrical, BPS, & Mechanical

Station Grade: Green = Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
Yellow = Needs Improvement (Dynamics Ready)
Red = Fail / Safety Hazard

TEAM:	**#:**

Regulation	**Grade**	**Comments**
3.1.B.1 **Safety Equipment** – Minimum requirements		
First Aid Kit (ANSI Z308.1 Class A or B, Type III or IV)		
ABC Fire Extinguisher –13.5 kg (30 lb) total		
Safety Vests – 1 per person		
Battery MSDS		
Spill Kit/method of containment of battery fires – 40 kg (88 lb) of sand		
Shovel / Spade – for applying sand		
Battery handling PPE – gloves, safety glasses, etc		
5 gal metal container for damaged electrochemical cells		
3.1.A **Safety Officer**		

3.1.A.1 **Team Safety Officer Name(s)**	3.1.A.2 **Proof of Training**		3.1.A.3 **Designated Safety Officer is not a Solar Car Driver, Solar Car Passenger, Support Vehicle Driver, or Team Manager**
	First Aid	**CPR**	

Sticker & Initials

Station Manager: ____________________

Entrance: Safety Officer(s) must be present

Station Grade: Pass (Track & Tour Ready)
Blue = *Not available at this station*
Yellow = *Not available at this station*
Red = Fail / Safety Hazard

TEAM:	**#:**

8.11 & 8.12.G **SOV/MOV Impound**	**Result/Comments**
Solution doesn't contain external hardware or removable hinge pins & allows main battery power connectors/conductors to be locked/sealed to prevent unmetered battery charging	☐Pass ☐Fail ☐Impound in Vehicle ☐External Impound
In vehicle solution for battery enclosure lid(s) & openings such as air inlet(s)/outlet(s)	☐Pass ☐Fail ☐N/A
In vehicle solution for motor, solar collector & other (describe ______) power port(s)	☐Pass ☐Fail ☐N/A
External impound box fits, fully contains & fully restricts access to battery enclosure	☐Pass ☐Fail ☐N/A
External impound box top has school name & team # in at least *10mm high letters*	☐Pass ☐Fail ☐N/A
Max 4 (in vehicle) or 2 (external) seals to secure/unsecure impound *(10 mm labels)*	☐Pass ☐Fail
Class: ☐SOV (Skip remaining inspections & proceed to grade station) ☐MOV (Proceed with inspections)	
8.12.A **MOV Charger**	**Result/Comments**
Onboard vehicle charger rigidly secured in vehicle	☐Pass ☐Fail
Charger protected from water ingress	☐Pass ☐Fail
Charger able to accept input voltages from 120-240 Vac	☐Pass ☐Fail
Charger power rating [kW]	
Describe DC charge current limiting methodology:	
▪ Considers max battery DC charge current limit from BMS	☐Yes ☐No ☐Unknown
▪ Considers the J1772 or NACS control pilot max AC current limit	☐Yes ☐No ☐Unknown
▪ Considers user set max charge rate	☐Yes ☐No ☐Unknown
8.12.B & 8.12.C **MOV Vehicle Power Inlet & MOV Charging Adapter**	**Result/Comments**
Standard EV power inlet receptacle present	☐Pass ☐Fail
Vehicle power inlet securely mounted to vehicle	☐Pass ☐Fail
Adapter needed for J1772 or NACS plug to another standard EV power inlet	☐Yes ☐No
▪ Charging adaptor isn't longer than 1m in length	☐Pass ☐Fail ☐N/A
▪ Charging adapter carried in vehicle when not in use	☐Pass ☐Fail ☐N/A
8.12.D **MOV Energy Metering**	**Result/Comments**
Sealed IEF energy meter assigned to team (Meter # __________________) (Seal # __________________)	
Plug the IEF onboard energy meter into the NEMA 14-50 inline connection	☐Pass ☐Fail
Energy meter display location is can be visibly read while charging	☐Pass ☐Fail
Charger is sealed to prevent unauthorized internal access	☐Pass ☐Fail
Battery enclosure features dedicated charger power port & contactor	☐Pass ☐Fail
8.12.E **MOV Charging Safety**	**Result/Comments**
BPS actively monitors/protects the battery during charge	☐Pass ☐Fail
When Main Power Contactors open for BPS fault, the charger contactor also opens	☐Pass ☐Fail
Charge current is automatically limited as battery nears full charge to avoid faults	☐Yes No ☐Unknown
AC/DC power connection enclosures/covers	
▪ Non-conductive	☐Pass ☐Fail
▪ Only removable with the use of tools	☐Pass ☐Fail
▪ 10 mm high letters with "Caution: High Voltage"	☐Pass ☐Fail
Power conductors sized appropriately for max AC/DC currents	☐Pass ☐Fail
▪ AC power min conductor size [AWG]	
▪ DC power min conductor size [AWG]	

TEAM:	**#:**

8.12.F **MOV Electric Vehicle Supply Equipment (EVSE)**	**Result/Comments**
Team has J1772 or NACS EVSE	☐J1772 ☐NACS ☐Fail
EVSE can be plugged into standard NEMA 5-15 120Vac outlet	☐Yes ☐No
120Vac compatible (list max AC current [A]: _________)	☐Yes ☐No
240Vac compatible (list max AC current [A]: _________)	☐Yes ☐No
Team has a generator that can be used to charge the vehicle	☐Yes ☐No ☐Unknown
MOV Charging Testing/Metered Charging Certification	**Result/Comments**
With EVSE disconnected & vehicle powered off, verify pinout at a slightly separated NEMA 14-50 connection	
▪ Neutral (W) - no continuity with other terminals	☐Pass ☐Fail
▪ Ground (G) - continuity with vehicle power inlet GND & any exposed charger/chassis metal but no other terminals	☐Pass ☐Fail
▪ L1 (B) - continuity with a vehicle power inlet Line but no other terminals	☐Pass ☐Fail
▪ L2 (R) - continuity with a vehicle power inlet Line but no other terminals	☐Pass ☐Fail
Verify charger power conductor isolation	
▪ AC input power to DC output power conductors	☐Pass ☐Fail
▪ AC input power conductors to vehicle chassis	☐Pass ☐Fail
▪ DC output power conductors to vehicle chassis	☐Pass ☐Fail
Have the team demonstrate charging with their own J1772 or NACS EVSE	☐Pass ☐Fail
Have the team demonstrate charging with IEF J1772 or NACS EVSE	☐Pass ☐Fail ☐Untested
▪ Proximity Pilot Validation	☐Pass ☐Fail ☐Untested
▪ Control Pilot Validation	☐Pass ☐Fail ☐Untested
Verify the IEF onboard energy meter is reading correctly	☐Pass ☐Fail
Vehicle drive motor is disabled when a J1772 or NACS plug is connected	☐Pass ☐Fail
Vehicle charging system is able to detect a broken AC ground scenario	☐Yes ☐No ☐Untested
Inspect & lock/seal all exposed connectors/conductors on the AC/DC charging power lines between this battery enclosure port & the vehicle power inlet to physically prevent any of these connections from being unplugged or tapped into	☐Pass ☐Fail
Energy Storage Pack Capacity (Q) from Electrical Station [kWh]	
Vehicle certification for metered charging in this event	☐Certified ☐Not Certified

Sticker & Initials

Station Manager:

Entrance: **External Impound Teams:** Battery Enclosure & Impound Box
Impound In Chassis Teams: Fully assembled solar car & Yellow status or better in Electrical & BPS
MOV Teams: Same requirements as Impound In Chassis + EVSE & Battery Spill Kit

Station Grade: Pass (Track & Tour Ready)
Blue = Pass with Penalty Condition (Track Ready)
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For Impound In Vehicle Teams, battery seals need to be verified before FSGP and photo documentation created for ASC Impound

TEAM:	**#:**

Regulation	**Lead**	**Chase**	**Scout**	**T&T**			**Comments**
12.5, 12.5.A – 12.5.E **Support Vehicles**	TOTAL # OF SUPPORT VEHICLES: __________						
All vehicles registered with Event HQ							
Not larger than 15 passenger full size van							
Roof mounted flashing amber lights							
GPS for observer viewing							
Storage racks are secure and safe							
12.5.F **Support Vehicle Graphics**							
Host institution name prominent on both sides							
Team number on both sides & rear (at least 250 mm high, with 40 mm brush stroke)							
Team number on top passenger's side of windshield (at least 150 mm high)							
Event logo on both sides of each vehicle and trailer							
Solar car caravan sign							
12.6 **Radio Communication**							
Communication with solar car driver, which observer can monitor							
Hands free comm. for all vehicle drivers							
Separate CB channel for ASC communications in all vehicles on route							
3.1.B.2, 12.5.B – 12.5.C **Safety Equipment** – Minimum requirements							
4 Orange Cones – minimum 300mm (12") high							
Orange Warning Flag							
First aid kit, fire extinguisher (sufficient capacity), safety vests (1 per person in vehicle)							
Battery spill kit: Battery MSDS, 40kg sand, shovel, PPE, suitable container(s) for damaged cells							
Demonstrations							
Roadside safety procedures by team (role play)							
CB radio check at range							

Sticker & Initials

Station Manager: ______________________________

Entrance: All support vehicles/equipment, team members who will be in lead & chase, and safety Officer(s);
Green status in Safety

Station Grade: Pass (Track & Tour Ready)
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